UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	:	CSC429
EXAMINATION	:	NOVEMBER 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC429 mock paper (November 2026). All answers are computed deterministically. Worked answers are shown for data representation and assembly; descriptive marking points come from a fixed bank.

2. No placeholders are used; every slot has a worked or descriptive answer.

3. Part A = 10 questions (80 marks); Part B = 1 assembly question (20 marks); total = 100 marks.

4. Computed slots: 4; descriptive slots: 7; placeholders: 0. Questions and answers are for local study use only.

5. Fingerprint based on 20 unique CSC429 papers; output is deterministic for seed 19.

FOR MARKERS' USE ONLY

This marking scheme consists of 9 printed pages.

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PART A — MARKING SCHEME

Worked answers and marking points.

QUESTION A1 (5 marks)

Study Figure 1 and answer the following questions.

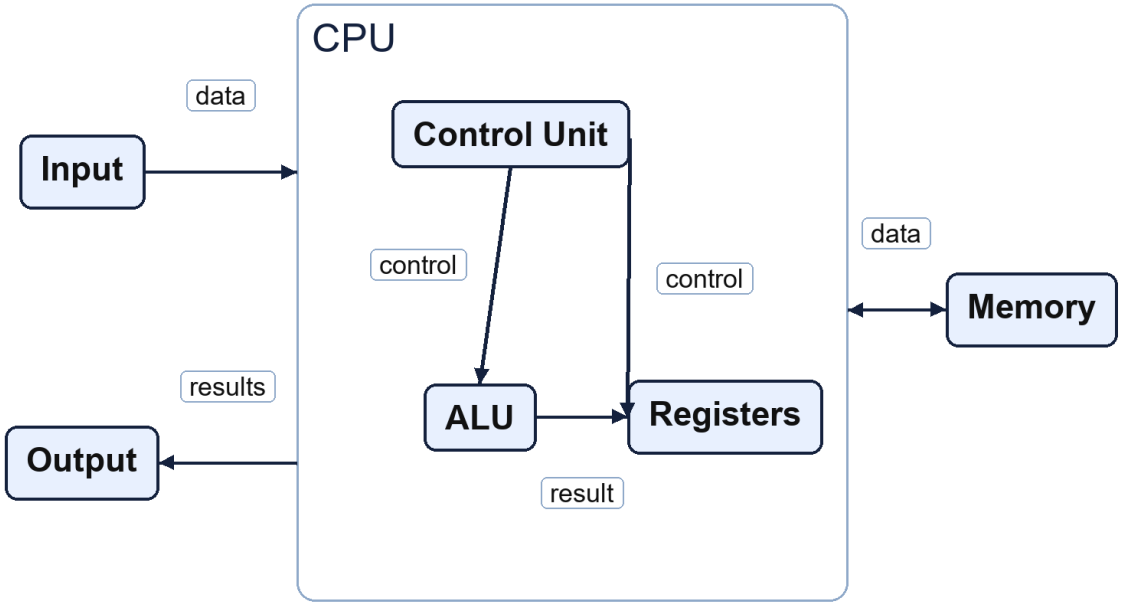


Figure 1: Stored-program computer organization.

(a) (2 marks) Name the computer organization shown in the figure and state its two key principles.

Answer: Von Neumann architecture. Instructions and data are stored in the same memory, and the CPU executes instructions sequentially unless a branch changes the control flow.

(b) (3 marks) Explain the function of the control unit and the arithmetic logic unit in this organization.

Answer: The control unit fetches and decodes instructions and generates the control signals. The ALU performs arithmetic and logical operations and records status flags.

QUESTION A2 (5 marks)

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The integrated circuit was an important milestone in computer history.

(a) (2 marks) Define an integrated circuit.

Answer: A small semiconductor chip containing many transistors and other electronic components fabricated together.

(b) (3 marks) State THREE advantages of integrated circuits over discrete transistors.

Answer: Smaller size, lower power consumption, higher speed/reliability, and lower cost in large quantities.

QUESTION A3 **(5 marks)**

Answer the following data-representation questions. Show your complete workflow.

(a) (2 marks) Convert 210 to binary. Show your workflow.

Answer: 210 = 11010010_2.

(b) (3 marks) Convert 11100110_2 to decimal. Show your workflow.

Answer: $11100110_2 = 230_{10}$.

QUESTION A4 **(5 marks)**

Answer the following data-representation questions. Show your complete workflow.

(a) (2 marks) Represent -66 in 8-bit sign-magnitude form.

Answer: 11000010 (sign bit 1, magnitude 66).

(b) (3 marks) Represent -66 in 8-bit two's complement form.

Answer: 10111110.

QUESTION A5 **(10 marks)**

Answer the following data-representation questions. Show your complete workflow.

(a) (2 marks) Convert 116 to binary.

Answer: 116 = 01110100_2.

(b) (2 marks) Convert 01101000_2 to hexadecimal.

Answer: $01101000_2 = 68_{16}$.

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(c) (3 marks) Convert 97_{16} to octal.

Answer: $97_{16} = 227_{8}$.

(d) (3 marks) Convert 132_{8} to decimal.

Answer: $132_{8} = 90_{10}$.

QUESTION A6 **(10 marks)**

A computer uses system interconnection and CPU interconnection.

(a) (3 marks) Differentiate system interconnection and CPU interconnection.

Answer: System interconnection connects the CPU, memory and I/O devices; CPU interconnection connects components inside the processor, such as cores and cache.

(b) (4 marks) Describe the address bus, data bus and control bus.

Answer: Address bus carries the address; data bus carries data/instructions; control bus carries control and timing signals.

(c) (3 marks) Explain how bus width and bus timing affect performance.

Answer: A wider data bus transfers more bits per cycle; faster timing/clocking reduces the time per transfer, both increasing bandwidth.

QUESTION A7 **(10 marks)**

Study Figure 2 and answer the following questions.

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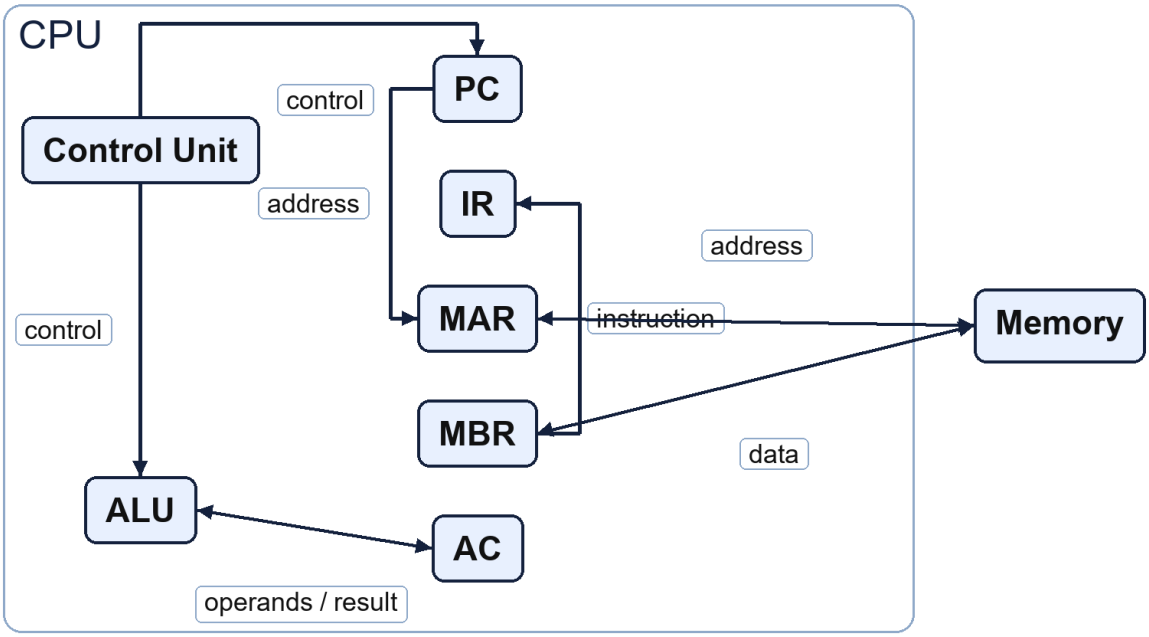


Figure 2: CPU, memory and system bus organization.

(a) (2 marks) Identify the register that holds the address of the next instruction and the register that holds the instruction currently being decoded.

Answer: PC holds the address of the next instruction; IR holds the instruction currently being decoded.

(b) (3 marks) Explain the roles of MAR and MBR during a memory read.

Answer: The MAR holds the address to be accessed. The memory places the selected word in the MBR, which then supplies it to the CPU (for example, to the IR).

(c) (5 marks) Describe the fetch-decode-execute cycle using the components shown in the figure.

Answer: Fetch: PC -> MAR; memory read; MBR -> IR; PC is incremented. Decode: the CU interprets the opcode in IR and issues control signals. Execute: operands are read, the ALU performs the operation, the result and flags are written back, and the next instruction is fetched.

QUESTION A8 **(10 marks)**

Study Figure 3 and answer the following questions.

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Tag (4 bits)	Line (3 bits)	Offset (9 bits)

Cache state

Line	Valid	Tag	Data
0	0	----	----
1	0	----	----
2	0	----	----
3	1	0010	0xA5
4	0	----	----
5	1	0001	0x3C
6	0	----	----
7	0	----	----

Addresses
0x1A2B
0x2A2B

Figure 3: Direct-mapped cache organization and address fields.

(a) (2 marks) The cache uses a 16-bit address split into a 4-bit tag, a 3-bit line number and a 9-bit offset. For address 0x1A2B, give the tag, line number and offset in binary.

Answer: 0x1A2B = 0001 1010 0010 1011; tag = 0001, line = 101, offset = 000101011.

(b) (4 marks) Using the cache state shown in the figure, determine whether each address is a hit or a miss: (i) 0x1A2B, (ii) 0x2A2B. Explain your reasoning.

Answer: (i) Hit: tag 0001 and line 101; line 5 is valid with tag 0001. (ii) Miss: tag 0010 and line 101; line 5 is valid but its tag is 0001, so the tags do not match.

(c) (4 marks) Explain why the cache compares tags and how temporal and spatial locality improve the hit rate.

Answer: The tag identifies which memory block occupies a line; comparing tags prevents returning data from a different block. Temporal locality means recently used data is likely reused, and spatial locality means nearby addresses are likely used, so keeping blocks in cache raises the hit rate.

QUESTION A9 **(10 marks)**

Interrupts can be maskable or non-maskable.

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(a) (3 marks) Differentiate maskable and non-maskable interrupts.

Answer: Maskable interrupts can be disabled/delayed by the CPU; non-maskable interrupts cannot be ignored and are used for critical events.

(b) (3 marks) Explain what an interrupt handler (ISR) does.

Answer: It saves context, identifies the device/cause, services the request, clears the interrupt and restores context.

(c) (4 marks) Describe TWO methods for handling multiple interrupts.

Answer: Priority-based nesting and polling/sequential service; nested interrupts allow a higher-priority request to pre-empt a lower-priority ISR.

QUESTION A10 **(10 marks)**

Multicore processors are common in modern computers.

(a) (2 marks) Define multicore processing.

Answer: A single processor package contains multiple independent cores that can execute instructions concurrently.

(b) (4 marks) Explain TWO benefits of multicore processing.

Answer: Higher throughput on parallel workloads and better performance per watt; it can also improve responsiveness by running tasks concurrently.

(c) (4 marks) Describe NUMA and ONE challenge in a distributed-memory system.

Answer: NUMA: memory access time depends on which node owns the data. Challenge: inter-node communication/synchronisation overhead and programming complexity.

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PART B — MARKING SCHEME

Worked answers and marking points.

QUESTION B1 **(20 marks)**

Write appropriate programs using zero-, one- and three-address instructions to compute Z = (A + B) * (C - D) / E. Registers A, B, C, D and E must not be modified; X, Y and Z are temporary registers; the final answer must be stored in Z. Available instructions: zero-address PUSH, POP, ADD, SUB, MUL, DIV; one-address LOAD, STORE, ADD, SUB, MUL, DIV; three-address ADD, SUB, MUL, DIV.

(a) (6 marks) Write a zero-address program for the expression.

Answer:

```
PUSH A
PUSH B
ADD
PUSH C
PUSH D
SUB
MUL
PUSH E
DIV
POP Z
```

(b) (6 marks) Write a one-address program for the expression.

Answer:

```
LOAD A
ADD B
STORE X
LOAD C
SUB D
STORE Y
LOAD X
MUL Y
STORE X
LOAD X
DIV E
STORE Z
```

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(c) (8 marks) Write a three-address program for the expression.

Answer:

```
ADD X, A, B
SUB Y, C, D
MUL X, X, Y
DIV Z, X, E
```

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